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Excess mortality related to the August 2003 heat wave in France

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1Epidémiologie environnementale des cancers

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Abstract

Objectives

From August 1st to 20th, 2003, the mean maximum temperature in France exceeded the seasonal norm by 11 to 12°C on nine consecutive days. The use of mortality data as a measure of the impact of the heat wave on the population was then observed, which main epidemiological features are described herein.

Methods

The number of deaths observed from August to November, 2003 in France was compared to those expected on the basis of the mortality rates in 2000 to 2002 and the 2003 population estimates.

Results

From August 1st to 20th, 2003, 15000 excess deaths were observed. From 35 years age, the excess mortality was marked and increased with age, being % higher in women than in men of comparable age as of age 45 years. Excess mortality at home and in retirement institutions was greater than in hospitals. The mortality of widowed, single and divorced subjects was greater than that of married people. Deaths directly related to stroke, hyperthermia and dehydration increased massively. Cardiovascular diseases, ill-defined morbid disorders, respiratory diseases and infectious diseases also markedly contributed to the excess mortality. The geographic variations in mortality showed a clear age-dependent relationship with the number of very hot days. No harvesting effect was observed.

Conclusions

Heat waves must be considered as a threat to European populations living in climates that are currently temperate. While the elderly and the very young are particularly vulnerable to heat waves, no segment of the population may be considered protected from the risks associated with heat waves.

Keywords: Adolescent; Adult; Age Factors; Aged; Aged, 80 and over; Cause of Death; Child; Child, Preschool; Female; France; epidemiology; Heat Wave; Heat effects; Heat Stress Disorders; mortality; Humans; Infant; Male; Marital Status; Middle Aged; Poisson Distribution; Seasons; Sex Factors; Temperature; Univariate Analysis

Keywords: Heat wave, Mortality, Modifying Factors, Causes of death, Harvesting

INTRODUCTION

From August 1st to 5th, 2003, the average maximum temperatures recorded in France increased from a value close to the normal value (25°C) to a value that remained between 36 and 37°C until August 13th, before beginning to fall (28°C on August 16th). Almost all of the population of France, approximately 60 million people, was exposed to the heat wave: the temperature exceeded 35°C for at least 9 days in 61 of the 96 French departments. A few days after the beginning of the heat wave, information reported by the medical emergency services, fire department and hospital emergency services showed an abrupt increase in the number of emergency interventions and in the mortality accompanying the heat wave.

The mortality associated with heat has been described and reviewed by several authors. [3,4]

Some authors have described the association between the usual daily fluctuations in temperature and mortality using time series or a space-time model. [6,9,10,26,28,31,33,37,40,44,46,47]

Others have focused on individual major heat episodes, in which an exceptional increase in mortality was observed concomitantly with an extreme heat wave. Those authors described the characteristics of the mortality during or after the heat wave. [1,6,9,11,12,19,25,28,35,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100] The probable increase in the frequency of such events in the 21st century has been stressed, as have the time-and-space variations of the main characteristics of that mortality.

Given that context and using the latter approach, the epidemiological characteristics of the excess mortality associated with the heat wave in August 2003 have been reported herein, on the basis of the studies carried out at the French National Institute for Health and Medical Research (Inserm) by the authors and publicly reported to the French health authorities. [17,18]

MATERIAL AND METHODS

The observations made from August 1st to November 30th, 2003 in mainland France, were compared to the similar observations for the period March to June 2000 to 2002, taken as the reference period. The mortality data were derived from the merged data bases of Inserm and INSEE (National Institute of Statistics and Economic Studies), thus ensuring that the data were complete.

A prior descriptive analysis of the daily mortality-rate process over the reference period has shown that, when the seasonal effect and trend were controlled, the residuals were distributed as a first-order autoregressive process with extra-Poisson variability. Two methods were used to estimate the expected number of deaths and the expected variance of daily death counts in 2003. Method A combined modelling of the monthly mortality rates, by age and gender, from 2000 to 2002, extrapolated to 2003, with estimations of the populations by year, from 2000 to 2003. The model adopted was a log-linear Poisson model incorporating a linear annual trend and a specific term for each year. Method B consisted in estimating the mean mortality rates by age, gender and month, from 2000 to 2002, and applying them to the 2003 population estimates. In both cases, an over-dispersion parameter was estimated and applied to parameter variance estimation. Ten-year age groups were considered.

For France as a whole, method A was adopted, since the daily numbers of deaths observed from March to June 2003 were, on average, close to the expected values calculated using method A, did not differ significantly from them, and differed significantly from the estimated values calculated using method B. At the finer geographical scale of the 22 "regions" and 96 "departements" that constitute mainland France, method B proved to be more appropriate. Method A on the basis of the same criteria.

Given the observed (O) and the expected (E) numbers of deaths, mortality was quantified using excess mortality (O-E), rounded up to the nearest integer, and mortality ratios (O/E). On the basis of the fluctuations in the daily numbers of deaths that were observed during the reference period of August in 2000, 2001 and 2002, "95% fluctuation intervals" were defined as intervals in which there was a 95% probability of observing a given number of deaths in the future if they had the same expected value, variance and auto-correlation as the deaths which were observed during the reference period. "Fluctuation intervals" are therefore prediction intervals.

The 95% fluctuation intervals of the observed mortality around the expected mortality for the period 2000-2002 were estimated by day, by region and month, taking into account the autoregressive structure of the daily mortality-rate process and the extra-Poisson variability observed during the reference period.

The analysis of the causes of death was conducted on the causes of death reported by the physicians making out the death certificates at the time of death. The latter is defined as the cause initiating the morbid process resulting in death. The causes of death were classified using the International Classification of Diseases (ICD10) and an additional category for deaths reported to be directly related to the heat wave by the physician: dehydration, hyperthermia and heatstroke.

Usually, coding the medical causes of death involves a long validation process. Validation is still ongoing for 2002. In the 2003 heat wave, validation was accelerated for August 2003. It was thus possible to compare the number of deaths by cause occurring in France in the first 10 days of August 2003 with the means of the same death counts for August 2000 and 2001, considered to be the expected number of deaths by cause. The statistical significance of the difference between the observed and expected number of deaths due to a given cause was determined under the hypothesis that the numbers of deaths used in the comparisons were independently Poisson distributed.

The daily maximum and minimum temperature statistics used are those derived from the Météo-France network of 192 meteorological stations covering all of the cities of France. For each of the 96 departements of France, a "number of very hot days" was determined. The number was defined as the number of days on which the minimum and maximum temperatures reported by the departement's meteorological stations simultaneously exceeded the corresponding 30-year averages by 5 and 9°C, respectively, between August 1st and 20th, 2003. The temperature cut-off points (5 and 9°C) were chosen to minimize the deviance of the relationship between the mortality ratios and the number of very hot days using a Poisson regression. The departements were divided into four groups with nearly equal numbers of expected deaths and an increasing number of very hot days: 0 to 3 days; 4 to 7 days; 8 to 13 days.

RESULTS

Time course of the heat wave and excess mortality

For the period from March to June 2003, approximately 95% of the observed mortality ratios were shown to lie within the 95% fluctuation intervals. This was also true for the daily time course of the mortality in July (figure 1).

Figure 1. Number of excess deaths observed during the French heat wave from July to September 2003 and average daily maximum (Tmax) and minimum (Tmin) temperatures recorded during the period.

Open in a new tab

The X axis shows the 92 days from July 1st to September 30th 2003. The histogram shows the numerical values of the daily excess mortality ratios, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.

The dotted horizontal lines show the limits of the 95% fluctuation intervals of the daily number of deaths.

In August, the mortality first deviated significantly from its usual range of fluctuation on the 4th. The daily excess then increased progressively, reaching 1193 deaths on August 8th and 2200 deaths on August 12th (figure 1). Excess mortality then fell to 1943 deaths on August 13th, 1988 on August 14th. Mortality had returned to normal as of August 19th. The fall-off in excess mortality reflected the fall in temperature around August 20th. The cumulative excess mortality from August 1st to 20th was 14 729 deaths, equivalent to an excess mortality ratio of 1.19.

Excess mortality by age and gender

Overall, no mortality ratio significantly greater than unity was observed for subjects aged less than 35 years (table 1). From the age of 35 years, the mortality ratios were significantly greater than 1 and increased regularly with age, rising from 1.19 for subjects aged 35 to 44 years to 1.98 for subjects aged 95 years and over. The number of excess deaths increased very rapidly with age: 11 731 of the 14 729 excess deaths occurred in subjects aged 65 years and over.

ts aged 75 years and over and 2930 in subjects aged 35 to 74 years.

Table 1.

Excess deaths by age and gender - France - August 1st to 20th, 2003.

Age ;
Total
;
Males
;
Females
;
;
Pop ;
O-E ;
O/E [CI(95%)] ;
Pop ;
O-E ;
O/E [CI(95%)] ;
Pop ;
O-E ;
O/E [CI(95%)] ;
;
< 35 years ;
26.9 ;
67 ;
1.1 [1.0; 1.2] ;
13.7 ;
73 ;
1.1 [1.0; 1.2] ;
13.3 ;
26 ;
1.0 [0.9; 1.1] ;
<1 year ;
0.7 ;
15 ;
1.1 [0.9; 1.3] ;
0.4 ;
25 ;
1.3 [1.0; 1.6] ;
0.4 ;
29 ;
0.9 [0.7; 1.1] ;
1-14 years ;
10.4 ;
4 ;
1.0 [0.8; 1.3] ;
5.3 ;
24 ;
0.9 [0.7; 1.3] ;
5.1 ;
8 ;
1.2 [0.8; 1.8] ;
15-24 years ;
7.8 ;
24 ;
1.1 [0.9; 1.3] ;
4.0 ;
26 ;
1.1 [1.0; 1.4] ;
3.8 ;
22 ;
1.0 [0.7; 1.3] ;
25-34 years ;
8.0 ;
24 ;
1.1 [0.9; 1.2] ;
4.0 ;
26 ;
1.1 [1.0; 1.3] ;
4.0 ;
22 ;
1.0 [0.8; 1.2] ;
35-74 years ;
28.2 ;
2930 ;
1.3 [1.3; 1.4] ;
13.7 ;
1773 ;
1.3 [1.2; 1.3] ;
14.5 ;
1157 ;
1.4 [1.3; 1.4] ;
35 -44 years ;
8.6 ;
151 ;
1.2 [1.1; 1.3] ;
4.3 ;
147 ;
1.3 [1.2; 1.4] ;
4.4 ;
4 ;
1.0 [0.9; 1.2] ;
45-54 years ;
8.3 ;
488 ;
1.3 [1.2; 1.3] ;

4.1 ;
 364 ;
 1.3 [1.2; 1.4] ;
 4.2 ;
 124 ;
 1.2 [1.1; 1.3] ;
 55-64 years ;
 6.2 ;
 615 ;
 1.3 [1.2; 1.3] ;
 3.1 ;
 357 ;
 1.2 [1.1; 1.3] ;
 3.2 ;
 258 ;
 1.4 [1.2; 1.5] ;
 65-74 years ;
 5.1 ;
 1676 ;
 1.4 [1.3; 1.5] ;
 2.3 ;
 905 ;
 1.3 [1.2; 1.4] ;
 2.8 ;
 771 ;
 1.5 [1.4; 1.6] ;
 ? 75 years
 ;
 4.7 ;
 11731 ;
 1.7 [1.6; 1.8] ;
 1.7 ;
 3505 ;
 1.5 [1.4; 1.6] ;
 3.0 ;
 8226 ;
 1.8 [1.7; 2.0] ;
 75-84 years ;
 3.6 ;
 4558 ;
 1.6 [1.5; 1.7] ;
 1.4 ;
 1910 ;
 1.5 [1.4; 1.5] ;
 2.2 ;
 2648 ;
 1.7 [1.6; 1.8] ;
 85-94 years ;
 1.0 ;
 5691 ;
 1.8 [1.7; 2.0] ;
 0.3 ;
 1355 ;
 1.6 [1.5; 1.7] ;
 0.7 ;
 4336 ;
 1.9 [1.8; 2.1] ;
 > ?95 years ;
 0.1 ;
 1482 ;
 2.0 [1.8; 2.2] ;
 0.0 ;
 240 ;
 1.8 [1.6; 2.1] ;
 0.1 ;
 1242 ;
 2.0 [1.8; 2.3] ;
 Total ;
 59.9 ;
 14729 ;
 1.5 [1.5; 1.6] ;
 29.1 ;
 5351 ;
 1.4 [1.3; 1.5] ;
 30.8 ;
 9378 ;
 1.7 [1.6; 1.8] ;
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 Pop: estimated population in June 2003 (millions inhabitants); O: observed number of deaths; E: expected number of deaths; O-E: excess
 up to the nearest integer; O/E: mortality ratio; CI(95%): 95% confidence interval of O/E.
 Among male subjects, significant excess mortality was observed for infants aged less than 1 year (+29%) and adults aged 35 to 44 years
 for female subjects in those age groups, mortality was normal. From age 55 years, the mortality ratios by age group were significantl
 men, by about 15%. The total excess mortality for women (+9378 deaths) was 75% higher than that for men (+5351 deaths).
 Excess mortality by place of death
 Major excess mortalities were observed for deaths at home (+5130 deaths), in retirement homes (+2574 deaths) and in public hospitals (+
 he relative increase in mortality at home (+74%) and in retirement homes (+91%) was, however, markedly greater than that observed in
 s (+45%) and private clinics (+22%). This was observed both globally and by age group (table 2).
 Table 2.
 Excess deaths by place of death, marital status, number of "very hot days" and age - France, August 1st to 20th, 2003.
 ;
 < 55 years
 ;
 55-74 years
 ;
 ? 75 years

```

;
All
;
;
0-E ;
0/E [CI(95%)] ;
0-E ;
0/E [CI(95%)] ;
0-E ;
0/E [CI(95%)] ;
0-E ;
0/E [CI(95%)] ;
;
Place of death ;
Home ;
343 ;
1.3 [1.3; 1.4] ;
951 ;
1.6 [1.5; 1.7] ;
3836 ;
1.9 [1.8; 2.0] ;
5130 ;
1.7 [1.6; 1.8] ;
Institutions, retirement homes ;
2 ;
1.2 [0.6; 2.1] ;
190 ;
1.9 [1.7; 2.2] ;
2382 ;
1.9 [1.7; 2.1] ;
2574 ;
1.9 [1.7; 2.1] ;
Public hospitals ;
239 ;
1.1 [1.1; 1.2] ;
1 038 ;
1.3 [1.2; 1.3] ;
4719 ;
1.6 [1.5; 1.7] ;
5996 ;
1.5 [1.4; 1.5] ;
Private hospitals and clinics ;
24 ;
1.1 [0.9; 1.3] ;
24 ;
1.0 [0.9; 1.1] ;
459 ;
1.4 [1.3; 1.5] ;
507 ;
1.2 [1.1; 1.3] ;
Street ;
29 ;
1.1 [1.0; 1.2] ;
9 ;
1.1 [0.9; 1.3] ;
77 ;
0.9 [0.7; 1.2] ;
30 ;
1.1 [1.0; 1.2] ;
;
Marital status ;
Single ;
434 ;
1.3 [1.2; 1.4] ;
645 ;
1.8 [1.7; 2.0] ;
1392 ;
2.1 [0.9; 3.6] ;
2471 ;
1.7 [1.6; 1.8] ;
Divorced ;
124 ;
1.2 [1.1; 1.4] ;
358 ;
1.5 [1.4; 1.6] ;
573 ;
2.0 [1.9; 2.3] ;
1055 ;
1.6 [1.5; 1.7] ;
Widowed ;
24 ;
1.4 [1.1; 1.8] ;
486 ;
1.5 [1.4; 1.6] ;
7186 ;
1.8 [1.4; 1.6] ;
7696 ;
1.8 [1.6; 1.9] ;
Married ;
211 ;
1.2 [1.1; 1.2] ;
834 ;
1.2 [1.1; 1.3] ;
2682 ;
1.5 [1.8; 2.2] ;

```

3727 ;
1.3 [1.3; 1.4] ;
;
Number of "very hot days" ;
0-1 day ;
55 ;
1.1 [1.0; 1.2] ;
201 ;
1.1 [1.0; 1.2] ;
1591 ;
1.4 [1.2; 1.5] ;
1848 ;
1.3 [1.2; 1.4] ;
2-3 days ;
125 ;
1.2 [1.1; 1.3] ;
316 ;
1.2 [1.1; 1.3] ;
1390 ;
1.4 [1.3; 1.5] ;
1830 ;
1.3 [1.2; 1.4] ;
4-7 days ;
265 ;
1.3 [1.2; 1.4] ;
694 ;
1.4 [1.3; 1.4] ;
3500 ;
1.8 [1.7; 1.9] ;
4459 ;
1.6 [1.5; 1.7] ;
? 8 days ;
260 ;
1.3 [1.2; 1.4] ;
1078 ;
1.6 [1.5; 1.7] ;
5249 ;
2.2 [2.1; 2.4] ;
6586 ;
2.0 [1.8; 2.1] ;
Open in a new tab

Only the deaths for which the place of death or the marital status was available have been included in the table. O: observed number of
ected number of deaths; O-E: excess deaths, rounded up to the nearest integer; O/E: mortality ratio; CI (95%): 95% confidence interval
Number of "very hot days": number of days on which the minimum and maximum temperatures have simultaneously exceeded the corresponding
e by respectively 5°C and 9°C between August 1st and 20th, 2003

Excess mortality by matrimonial status

The mortality ratios by age group over the period from August 1st to 20th (table 2) were much higher for single people than for married
divorced and widowed subjects showed intermediate values.

Medical causes of death

Certain medical causes of death had a major impact on the overall increase in mortality (table 3):

Table 3.

Excess deaths by medical cause and age - France - August 1st to 20th, 2003.

Causes of death (ICD10 Codes) ;

All

;

< 45 years

;

45-74 years

;

? 75 years

;

;

O-E ;

O/E [CI(95%)] ;

O-E ;

O/E [CI(95%)] ;

O-E ;

O/E [CI(95%)] ;

O-E ;

O/E [CI(95%)] ;

;

Dehydration, heatstroke, hyperthermia ;

3306 ;

21.3 [19.0; 23.9] ;

29 ;

- ;

425 ;

48.2 [45.8; 50.7] ;

2852 ;

19.6 [19.1; 20.1] ;

Dehydration (E86) ;

1628 ;

11.9 [10.5; 13.4] ;

7 ;

- ;

147 ;

20.5 [12.1; 34.8] ;

1475 ;

11.3 [10.0; 12.8] ;

Heatstroke (R509) ;

1313 ;

165.1 [100.8; 270.3] ;

15 ;

- ;

197 ;

198 [49.2; 797.3] ;
1101 ;
158.3 [93.4; 268.2] ;
Hyperthermia (X30) ;
365 ;
82.1 [42.4; 159.0] ;
7 ;
- ;
82 ;
164 [22.8; 1178.2] ;
276 ;
70 [34.7; 141.4] ;
Circulatory system diseases (I00 - I99) ;
3004 ;
1.4 [1.4; 1.4] ;
7 ;
1 [0.8; 1.2] ;
365 ;
1.2 [1.1; 1.3] ;
2633 ;
1.5 [1.5; 1.5] ;
Neoplasms (C00 - D48) ;
933 ;
1.1 [1.1; 1.1] ;
0 ;
1 [0.9; 1.1] ;
249 ;
1.1 [1.1; 1.1] ;
685 ;
1.2 [1.2; 1.2] ;
Respiratory system diseases (J00 - J99) ;
1365 ;
1.9 [1.8; 2.0] ;
4 ;
1.2 [0.7; 2.0] ;
149 ;
1.5 [1.3; 1.7] ;
1213 ;
2.1 [2.0; 2.2] ;
Nervous system diseases (G00 - H95) ;
1001 ;
2 [1.9; 2.1] ;
27 ;
1.5 [1.1; 2.0] ;
193 ;
1.7 [1.5; 1.9] ;
781 ;
2.1 [2.0; 2.3] ;
Mental disorders (F00 - F99) ;
748 ;
1.8 [1.7; 1.9] ;
12 ;
1.2 [0.9; 1.6] ;
187 ;
1.9 [1.7; 2.2] ;
550 ;
1.8 [1.7; 1.9] ;
Digestive system diseases (K00 - K93) ;
301 ;
1.2 [1.1; 1.3] ;
2 ;
1 [0.8; 1.3] ;
98 ;
1.2 [1.1; 1.3] ;
202 ;
1.3 [1.2; 1.4] ;
Endocrine and nutritional diseases1 (E00-E859.E87-E90) ;
620 ;
1.7 [1.6; 1.8] ;
3 ;
1.1 [0.7; 1.7] ;
143 ;
1.6 [1.4; 1.8] ;
475 ;
1.8 [1.7; 2.0] ;
Infectious diseases (A00 - B99) ;
483 ;
1.9 [1.7; 2.1] ;
720 ;
0.7 [0.5; 1.0] ;
91 ;
1.6 [1.3; 1.9] ;
413 ;
2.3 [2.1; 2.6] ;
Genitourinary system diseases (N00 - N99) ;
381 ;
1.9 [1.7; 2.1] ;
0 ;
1 [0.4; 2.7] ;
51 ;
1.7 [1.3; 2.2] ;
330 ;
2 [1.8; 2.2] ;
Musculoskeletal diseases (M00 - M99) ;
103 ;

1.6 [1.4; 1.9] ;
 7 ;
 4.5 [1.4; 14.6] ;
 20 ;
 1.5 [1.1; 2.1] ;
 76 ;
 1.5 [1.3; 1.8] ;
 Blood diseases (D50 - D89) ;
 55 ;
 1.5 [1.2; 1.8] ;
 72 ;
 0.8 [0.3; 1.9] ;
 16 ;
 1.7 [1.1; 2.6] ;
 41 ;
 1.6 [1.2; 2.1] ;
 Skin diseases (L00 - L99) ;
 48 ;
 1.4 [1.1; 1.7] ;
 0 ;
 - ;
 7 ;
 1.9 [0.9; 3.8] ;
 41 ;
 1.4 [1.1; 1.7] ;
 Congenital malformations (Q00 - Q99) ;
 14 ;
 1.2 [0.9; 1.5] ;
 1 ;
 1.0 [0.7; 1.4] ;
 19 ;
 2.1 [1.3; 3.4] ;
 75 ;
 0.5 [0.2; 1.2] ;
 Perinatal diseases (P00 - P96) ;
 6 ;
 1.1 [0.8; 1.4] ;
 6 ;
 1.1 [0.8; 1.4] ;
 0 ;
 - ;
 0 ;
 - ;
 Pregnancy. Childbirth (000 - 099) ;
 74 ;
 0 [0.0; 0.0] ;
 74 ;
 0.0 [0.0; 0.0] ;
 0 ;
 - ;
 0 ;
 - ;
 Injury, poisoning2 (V01 -X29. X31 - Y89) ;
 437 ;
 1.2 [1.1; 1.3] ;
 745 ;
 0.9 [0.8; 1.0] ;
 137 ;
 1.2 [1.1; 1.3] ;
 345 ;
 1.4 [1.3; 1.5] ;
 Ill-defined conditions3 (R00 - R508. R51 - R99) ;
 1741 ;
 2 [1.9; 2.1] ;
 59 ;
 1.3 [1.1; 1.5] ;
 418 ;
 2.1 [1.9; 2.3] ;
 1 265 ;
 2.1 [2.0; 2.2] ;
 ;
 Total ;
 14539 ;
 1.5 [1.5; 1.5] ;
 83 ;
 1.0 [0.9; 1.1] ;
 2565 ;
 1.3 [1.3; 1.3] ;
 11891 ;
 1.7 [1.7; 1.7] ;
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 1Excluding dehydration;
 2Excluding heatstroke;
 3Excluding hyperthermia;
 0: observed number of deaths, E: expected number of deaths, O-E: excess deaths, rounded up to the nearest integer; O/E: mortality ratio
 y ratio (noted " - " when E = 0); CI(95%): 95% confidence interval.
 - the causes directly related to the heat wave (heatstroke, hyperthermia and dehydration) responsible for 3306 deaths, i.e. a twenty-fo
 ease in the number of deaths, and an even greater increase in certain age-group categories,
 - circulatory system diseases (+3004 deaths), ill-defined morbid conditions (+1741 deaths), respiratory system diseases (+1365 deaths)
 tem diseases (+1001 deaths).
 Other medical causes of death, although contributing less markedly to the overall excess mortality, increased in frequency by 70 to 90%
 e period: mental disorders (+748 deaths), infectious diseases (+483 deaths), genitourinary system diseases (+381 deaths) and endocrin
 al diseases (+620 deaths).
 Almost all the other medical causes of death increased, but to a lesser extent.
 In subjects aged less than 45 years, only the causes of death directly related to the heat wave and death due to ill-defined morbid con

culoskeletal diseases increased. These increases were observed for men only. For subjects aged 45–74 years and those aged 75 years or older, the profile of excess mortality was similar and also similar to that for the total population.

Geographic variability of the excess mortality

In all the 22 regions of France a significant increase in mortality was observed from August 1st to 20th, 2003. However, the geographic distribution of the mortality was marked (figure 2). The highest excess mortality was observed in the Paris region (+142%), the most heavily populated area: +149% for the city of Paris and +174% for the nearest suburb (immediately surrounding Paris). The region contiguous with the Paris region to the south-west had the second highest mortality ratio (+104%), although it has no major cities and is relatively little urbanized. In the other regions, excess mortalities were observed in the Mediterranean, Atlantic and Channel coastal regions (figure 2 – left map). While the increases were statistically significant, they were smaller, of the order of 20%.

Figure 2. Geographic distributions of mortality ratios O/E (left map) and the number of “very hot days” (right map) by Département in France from August 1st to 20th, 2003.

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O: observed number of deaths; E: expected number of deaths; O/E: mortality ratio. The mortality ratios have been grouped into five classes: 0 to 1, 1 to 2, 2 to 3, 3 to 4, and 4 or more, approximately equal expected numbers of deaths.

The 96 French départements have been grouped into four classes on the basis of the number of “very hot days”. Number of “very hot days” is defined as the number of days on which the minimum and maximum temperatures simultaneously exceeded the corresponding 30-year averages by 5 and 9°C, respectively, from August 1st to 20th, 2003.

The mortality ratios observed over the period from August 1st to 20th in the 96 French départements increased with the number of very hot days (table 2). The increase was statistically significant for subjects aged less than 55 years, significant and very marked for subjects aged 55 to 74 years, significant and still more pronounced for subjects aged 75 years and older.

The geographic distribution of the number of “very hot days” (figure 2 – right map) was similar to that of mortality ratios (figure 2 – left map). With regard to infants aged less than 1 year, no excess mortality was observed for girls, either in the Paris region or in the rest of France. For the boys, the excess mortality in the Paris region (+55%) was significantly greater than that in the remainder of France (+12%).

Post-heat wave time course of overall mortality

The daily mortality observed from August 19th to September 30th, 2003, for the overall population did not deviate from its usual fluctuations (figure 1). Such was also the case for the daily mortality observed from October 1st to November 30th, 2003, for the overall population.

The return of mortality to normal was also observed on the ten-day period scale as of the third ten-day period in August, and, on the overall scale, as of the month of September.

Normalization affected both the overall population of France and the various sub-populations that exhibited various degrees of excess mortality. Mortality returned to normal as of August 19th and remained so until November 30th, 2003:

- for all age groups,
- for both men and women,
- for all place-of-death categories (at home, hospital, institutions and retirement homes),
- for all matrimonial status categories (single, divorced, widowed and married),
- for each of the 22 French regions and each of the 4 groups of départements defined by the number of very hot days experienced between August 1st and 20th, 2003.

DISCUSSION

In the first fortnight of August 2003, the entire population of the 60 million inhabitants of France was exposed to a heat wave of unprecedented intensity. The first significant excess mortality was observed 3 days after the maximum temperatures rose above the seasonal normal value. On successive days that maximum temperatures remained 11 to 12°C higher than the seasonal average, the daily excess mortality increased continuously. Mortality returned to normal for the overall population of France on August 19th, 4 days after the maximum temperatures had returned to values close to the seasonal averages. Almost 15 000 excess deaths occurred between August 1st and 20th, 2003, i.e. 55% more than the expected mortality. On a larger scale, from June to August 2003, the whole of Western Europe was struck by a series of heat waves. Excess mortality was observed in France (+1316 deaths), [36] Spain (+6595 to 8648 deaths), [45] Italy (in 21 cities: +3134 deaths), [8] England and Wales (+2139 deaths), [24] Germany (+1400 to 2200 deaths), [13] and Switzerland (+975 deaths). [15] Therefore, the overall excess death count of the 2003 heat wave is higher than that of any other European countries are considered from June to August. A large fraction of the excess deaths appears to have occurred in France. Between August 1st and 20th, with a marked increase in mortality applied to a large population (60 million people).

Numerous other episodes of major excess mortality associated with heat waves have been reported in North America, [1,6,11,19,25,35,42,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144,145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168,169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192,193,194,195,196,197,198,199,200,201,202,203,204,205,206,207,208,209,210,211,212,213,214,215,216,217,218,219,220,221,222,223,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,239,240,241,242,243,244,245,246,247,248,249,250,251,252,253,254,255,256,257,258,259,260,261,262,263,264,265,266,267,268,269,270,271,272,273,274,275,276,277,278,279,280,281,282,283,284,285,286,287,288,289,290,291,292,293,294,295,296,297,298,299,300,301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,322,323,324,325,326,327,328,329,330,331,332,333,334,335,336,337,338,339,340,341,342,343,344,345,346,347,348,349,350,351,352,353,354,355,356,357,358,359,360,361,362,363,364,365,366,367,368,369,370,371,372,373,374,375,376,377,378,379,380,381,382,383,384,385,386,387,388,389,390,391,392,393,394,395,396,397,398,399,400,401,402,403,404,405,406,407,408,409,410,411,412,413,414,415,416,417,418,419,420,421,422,423,424,425,426,427,428,429,430,431,432,433,434,435,436,437,438,439,440,441,442,443,444,445,446,447,448,449,450,451,452,453,454,455,456,457,458,459,460,461,462,463,464,465,466,467,468,469,470,471,472,473,474,475,476,477,478,479,480,481,482,483,484,485,486,487,488,489,490,491,492,493,494,495,496,497,498,499,500,501,502,503,504,505,506,507,508,509,510,511,512,513,514,515,516,517,518,519,520,521,522,523,524,525,526,527,528,529,530,531,532,533,534,535,536,537,538,539,540,541,542,543,544,545,546,547,548,549,550,551,552,553,554,555,556,557,558,559,560,561,562,563,564,565,566,567,568,569,570,571,572,573,574,575,576,577,578,579,580,581,582,583,584,585,586,587,588,589,590,591,592,593,594,595,596,597,598,599,600,601,602,603,604,605,606,607,608,609,610,611,612,613,614,615,616,617,618,619,620,621,622,623,624,625,626,627,628,629,630,631,632,633,634,635,636,637,638,639,640,641,642,643,644,645,646,647,648,649,650,651,652,653,654,655,656,657,658,659,660,661,662,663,664,665,666,667,668,669,670,671,672,673,674,675,676,677,678,679,680,681,682,683,684,685,686,687,688,689,690,691,692,693,694,695,696,697,698,699,700,701,702,703,704,705,706,707,708,709,710,711,712,713,714,715,716,717,718,719,720,721,722,723,724,725,726,727,728,729,730,731,732,733,734,735,736,737,738,739,740,741,742,743,744,745,746,747,748,749,750,751,752,753,754,755,756,757,758,759,760,761,762,763,764,765,766,767,768,769,770,771,772,773,774,775,776,777,778,779,780,781,782,783,784,785,786,787,788,789,790,791,792,793,794,795,796,797,798,799,800,801,802,803,804,805,806,807,808,809,810,811,812,813,814,815,816,817,818,819,820,821,822,823,824,825,826,827,828,829,830,831,832,833,834,835,836,837,838,839,840,841,842,843,844,845,846,847,848,849,850,851,852,853,854,855,856,857,858,859,860,861,862,863,864,865,866,867,868,869,870,871,872,873,874,875,876,877,878,879,880,881,882,883,884,885,886,887,888,889,890,891,892,893,894,895,896,897,898,899,900,901,902,903,904,905,906,907,908,909,910,911,912,913,914,915,916,917,918,919,920,921,922,923,924,925,926,927,928,929,930,931,932,933,934,935,936,937,938,939,940,941,942,943,944,945,946,947,948,949,950,951,952,953,954,955,956,957,958,959,960,961,962,963,964,965,966,967,968,969,970,971,972,973,974,975,976,977,978,979,980,981,982,983,984,985,986,987,988,989,990,991,992,993,994,995,996,997,998,999,1000] As was the case in those reported heat waves, the excess mortality observed in France in August 2003 increased with the number of very hot days. It is, however, noteworthy that even though the majority of the excess mortality affected subjects aged 75 years and over, very marked excesses were nonetheless observed in all age groups from 35 to 74 years (+2930 deaths). Moderate but significant excess mortality also occurred among male infants aged less than 1 year (+25 deaths). If the elderly are a particularly vulnerable population, the belief that they are at risk of heat-related death would thus be erroneous.

Both women, [34,39] and men, [48] have been reported to have higher mortality ratios than the opposite gender. In this study, the overall mortality for women was 75% higher than that for men. However, most of this difference may be explained by the greater longevity of women. When age is considered, from age 55 years, there is only a female excess mortality of about 15%. While the difference needs to be elucidated, it is prudent to consider that only women are exposed to the risk of death in heat waves.

While the relative increase in mortality observed in hospitals and clinics was less than that for subjects at home or in retirement homes, the excess mortality in the public hospitals was nonetheless major. Thus, all the medical emergency services and emergency care management were subject to exceptionally intense demand. The particularly high mortality ratios observed among people living at home and the substantial excess mortality ratios observed for single and divorced subjects compared to married subjects point to the key role of isolation in vulnerability to heat waves. [29,35,43] People living alone at home are to be considered a sub-population exposed to a particularly high risk.

The deaths occurring during the heat waves described in the literature were related to a wide variety of diseases. [1,11,19,34,39,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144,145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168,169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192,193,194,195,196,197,198,199,200,201,202,203,204,205,206,207,208,209,210,211,212,213,214,215,216,217,218,219,220,221,222,223,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,239,240,241,242,243,244,245,246,247,248,249,250,251,252,253,254,255,256,257,258,259,260,261,262,263,264,265,266,267,268,269,270,271,272,273,274,275,276,277,278,279,280,281,282,283,284,285,286,287,288,289,290,291,292,293,294,295,296,297,298,299,300,301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,322,323,324,325,326,327,328,329,330,331,332,333,334,335,336,337,338,339,340,341,342,343,344,345,346,347,348,349,350,351,352,353,354,355,356,357,358,359,360,361,362,363,364,365,366,367,368,369,370,371,372,373,374,375,376,377,378,379,380,381,382,383,384,385,386,387,388,389,390,391,392,393,394,395,396,397,398,399,400,401,402,403,404,405,406,407,408,409,410,411,412,413,414,415,416,417,418,419,420,421,422,423,424,425,426,427,428,429,430,431,432,433,434,435,436,437,438,439,440,441,442,443,444,445,446,447,448,449,450,451,452,453,454,455,456,457,458,459,460,461,462,463,464,465,466,467,468,469,470,471,472,473,474,475,476,477,478,479,480,481,482,483,484,485,486,487,488,489,490,491,492,493,494,495,496,497,498,499,500,501,502,503,504,505,506,507,508,509,510,511,512,513,514,515,516,517,518,519,520,521,522,523,524,525,526,527,528,529,530,531,532,533,534,535,536,537,538,539,540,541,542,543,544,545,546,547,548,549,550,551,552,553,554,555,556,557,558,559,560,561,562,563,564,565,566,567,568,569,570,571,572,573,574,575,576,577,578,579,580,581,582,583,584,585,586,587,588,589,590,591,592,593,594,595,596,597,598,599,600,601,602,603,604,605,606,607,608,609,610,611,612,613,614,615,616,617,618,619,620,621,622,623,624,625,626,627,628,629,630,631,632,633,634,635,636,637,638,639,640,641,642,643,644,645,646,647,648,649,650,651,652,653,654,655,656,657,658,659,660,661,662,663,664,665,666,667,668,669,670,671,672,673,674,675,676,677,678,679,680,681,682,683,684,685,686,687,688,689,690,691,692,693,694,695,696,697,698,699,700,701,702,703,704,705,706,707,708,709,710,711,712,713,714,715,716,717,718,719,720,721,722,723,724,725,726,727,728,729,730,731,732,733,734,735,736,737,738,739,740,741,742,743,744,745,746,747,748,749,750,751,752,753,754,755,756,757,758,759,760,761,762,763,764,765,766,767,768,769,770,771,772,773,774,775,776,777,778,779,780,781,782,783,784,785,786,787,788,789,790,791,792,793,794,795,796,797,798,799,800,801,802,803,804,805,806,807,808,809,810,811,812,813,814,815,816,817,818,819,820,821,822,823,824,825,826,827,828,829,830,831,832,833,834,835,836,837,838,839,840,841,842,843,844,845,846,847,848,849,850,851,852,853,854,855,856,857,858,859,860,861,862,863,864,865,866,867,868,869,870,871,872,873,874,875,876,877,878,879,880,881,882,883,884,885,886,887,888,889,890,891,892,893,894,895,896,897,898,899,900,901,902,903,904,905,906,907,908,909,910,911,912,913,914,915,916,917,918,919,920,921,922,923,924,925,926,927,928,929,930,931,932,933,934,935,936,937,938,939,940,941,942,943,944,945,946,947,948,949,950,951,952,953,954,955,956,957,958,959,960,961,962,963,964,965,966,967,968,969,970,971,972,973,974,975,976,977,978,979,980,981,982,983,984,985,986,987,988,989,990,991,992,993,994,995,996,997,998,999,1000] The variety of medical causes of the excess deaths in France in August 2003, some reflect the very morbid process which resulted in death. Thus, for the deaths due to heatstroke, hyperthermia and dehydration, but also, probably, for a fraction, at least, of the deaths for other particularly circulatory and respiratory diseases. [1,11,19,20,34,39,42] Certain medical causes of death may also reflect special vulnerability to heat waves associated with a chronic disease. This was particularly the case for deaths related to mental disorders, nervous system diseases, [41] and endocrine diseases, [34,42] but also, probably, for other causes.

The excess mortality was substantially greater in the départements that had the greatest number of very hot days and more markedly in the most urbanized areas. However, the relationship is far from perfect: the meteorological indicator of heat-wave exposure used herein is, in fact, only a crude indicator. A finer classification of the 96 French départements on an ordinal scale of heat-wave exposure. An analysis of heat heterogeneity on a finer scale is required. In addition, several heat-wave characteristics appear to have a strong impact on mortality: an abrupt major increase in temperature, whose impact is immediate or very short term; [26] a succession of very hot days; [14,47] and the time during the summer at which the heat wave occurs. [10] The exact form of the impact is still far from being elucidated. Heat waves may be described using multiple meteorological indicators, such as minimum, maximum or apparent temperature and relative humidity. [12,31,37] The latter remained particularly low during the August 2003 heat wave and the daily meteorological data for that indicator were therefore not taken into account herein.

The excess mortality observed showed great geographic heterogeneity ranging from +20% in the least-affected coastal regions to +142% in the Paris region. The particularly marked excess mortality in the city of Paris and the immediately surrounding suburbs is probably related to the higher population density in large metropolitan areas reported by certain authors. [6,28] This observation is to be considered in the context of the heat island effect, [32,42,46] and the question of whether there is a synergy between prolonged high temperatures and socioeconomic characteristics. [19,25] Some studies also suggest a possible contribution of atmospheric pollution to the increase in the number of deaths during heat waves. [1,11,19,20,34,39,42] In August 2003, ozone and particulate matter concentrations were particularly high in France: the ozone concentration alert threshold was exceeded for at least one recording station for 12 consecutive days from August 2nd to 13th, 2003. Moreover, exceptionally high values were recorded in areas where the ozone concentration is generally low. [21]

The above factors could explain some of the differences between the geographic distributions of mortality ratios and the number of “very hot days”. A finer geographic analysis of temperatures and further investigation of the vulnerability factors related to socioeconomic characteristics are required.

Few observations relating to the existence of a harvesting effect following earlier heat waves are available. They show that a harvesting effect was limited, [5,9,16,26,40] or even non-existent. [20,39] In the present study, no harvesting effect was observed; the excess mortality of the August 2003 heat wave was not followed by a less-than-expected number of deaths until the end of 2003. Moreover, the excess mortality in the first ten days of August 2003 was not followed, from August 21st to November 30th, 2003, by persistent excess mortality.

According to the IPCC's climate change models, [23] average temperatures should increase by 1.4 to 5.8°C between 1990 and 2100. In 2100, the probability of a summer heat wave similar to that which struck western Europe in August 2003 could be greater than 50%. [41] Given that the development of systems to prevent the risks related to heat waves, combining long-term preventive measures with alert systems based on short-term forecasts, needs to be seriously considered even in countries which have, so far, been relatively spared by heat waves. Several pro-

ert systems have been set up in response to heat waves associated with marked excess mortality (<http://www.ci.mil.wi.us>). [27,35,43] Institut de Veille Sanitaire (InVS), in cooperation with Météo-France set up such a system in June 2004. [22]

The epidemiological knowledge that could orient such preventive actions and alert systems is currently far from complete, given that the duration of population exposure to heat waves and the population sensitivities thereto appear to be population- and time-dependent [3], modeling heat-mortality relationships and on the individual and collective vulnerability factors that modulate those relationships is required.

The findings reported herein show that while there are populations that are particularly vulnerable to heat waves, such as the elderly living alone, no segment of the population may be considered protected from the risks associated with heat waves.

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